

Homework 2

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2026-09-26

Problem 1

Suppose that x, y are distinct points in a metric space (X, d) and let $\epsilon = \frac{d(x, y)}{2}$.
Prove that $B_\epsilon(x)$ and $B_\epsilon(y)$ are disjoint

Solution. Assume $\exists x_0 \in B_\epsilon(x)$ and $y_0 \in B_\epsilon(y)$ such that $x_0 = y_0$.
Thus $d(x_0, x) = d(y_0, y) = d(x_0, y) = d(y_0, x) < \frac{d(x, y)}{2}$ and $d(x_0, y_0) = 0$.
Then using the triangle inequality

$$\begin{aligned} d(y_0, x) + d(y, y_0) &\leq d(x_0, y_0) + d(x_0, x) + d(y_0, y) \\ \frac{d(x, y)}{2} + \frac{d(x, y)}{2} &\leq d(x_0, x) + d(y_0, y) < \frac{d(x, y)}{2} + \frac{d(x, y)}{2} \end{aligned}$$

No such number $d(x_0, x) + d(y_0, y)$ exists, a contradiction □

Problem 2

Sketch the open ball $B_1^{d_\infty}((0, 0))$ in $\mathbb{R}^2$

Problem 3

Prove that a subset of a metric space is open iff it is a union of open balls

Solution. Let (X, d) be a metric space and $U \subset X$

($\Rightarrow$)

Assume U is open

Then $\forall x \in U \exists \epsilon_x > 0$ s.t. $B_{\epsilon_x} \subset U$

Then looking at $\cup_{x \in U} B_{\epsilon_x}(x)$

If $u \in U$, then certainly $u \in B_{\epsilon_u}(u)$, so $u \in \cup_{x \in U} B_{\epsilon_x}(x)$

If $u \in \cup_{x \in U} B_{\epsilon_x}(x)$, which is a subset of U since U is open so $u \in U$

Thus $U = \cup_{x \in U} B_{\epsilon_x}(x)$

($\Leftarrow$) □

Problem 4

Suppose that in a metric space X we have $B_r(x) = B_s(y)$ for some $x, y \in X$ and some positive real numbers r, s

Is $x = y$? and is $r = s$?

Solution. It is not true in general that $x = y$ and $r = s$

Consider the metric space of $(\mathbb{R}^n, d)$ for any n , where $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$

Now for any $r \geq 1$ we have that $B_r(x) = \mathbb{R}^n$ for all $x \in \mathbb{R}^n$

For $n = 1$, we can take $x = 1$ and $y = 2$, $r = 2$ and $s = 3$, we have $B_r(x) = B_s(y) = \mathbb{R}$ while $x \neq y$ and $r \neq s$ □

Problem 5

Which of the following sets are closed in $\mathbb{R}$?

- $[1, \infty)$
- $\mathbb{R} \setminus \mathbb{Q}$
- $\left\{ \frac{n}{n+1} \mid n \in \mathbb{N} \right\}$
- $\left\{ \frac{1}{n} \mid n \in \mathbb{N}, n \geq 2 \right\} \cup \{0, 1, 2\}$